



Pediatric Emergency Research Network



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**Sedation may not be perfect
But must be SAFE**

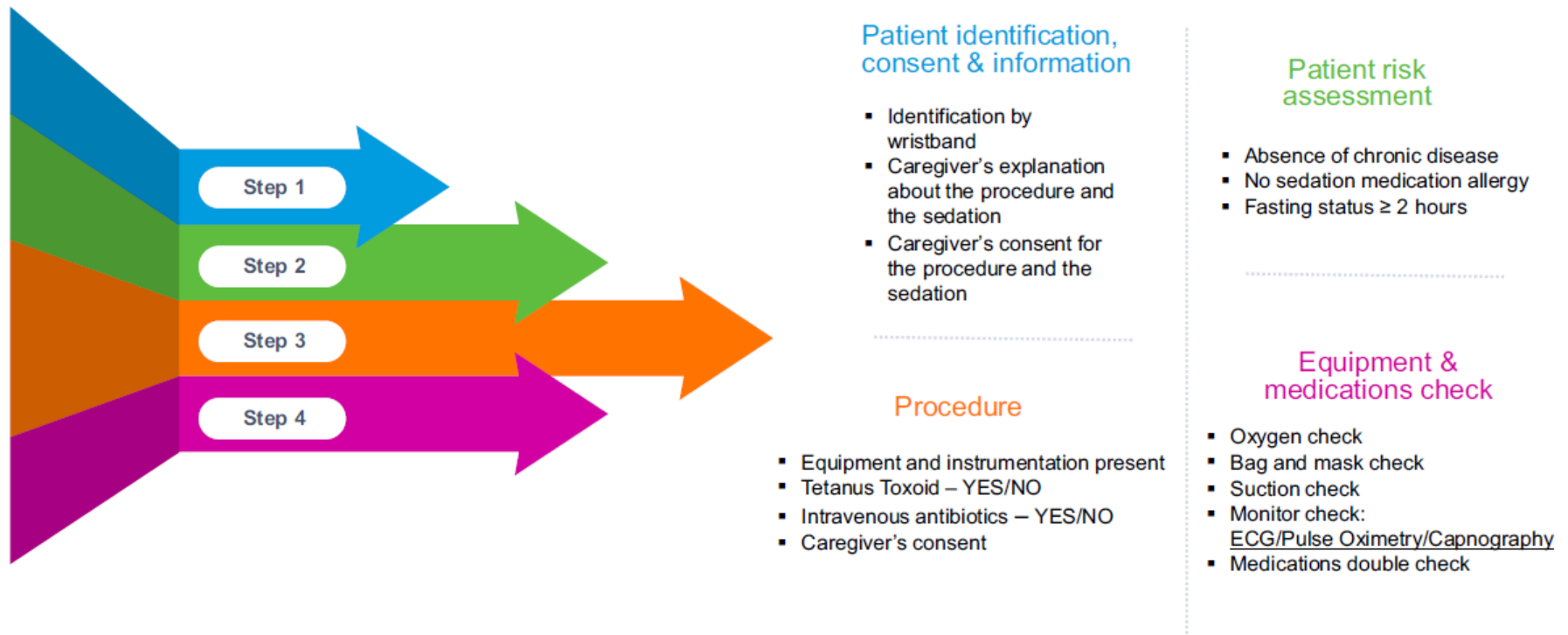
The Joint Commission Journal on Quality and Patient Safety

Table 4. Emergency Department Procedures, Risks, Safety Hazards, and Recommendations*

Emergency Department Procedure	General Risk Level of Procedure	Specific Safety Hazard [†]				Recommendations				
		Wrong Patient	Wrong Site	Medication Allergies/ Interactions	Equipment Issues (Inadequate/ Malfunctioning)	Emergency Procedure	Time-Out [‡]	Preprocedure Check	Equipment Check (by Type of Procedure)	Postprocedure Check
Procedural Sedation	High	X		X	X		X	X	X	X
Rapid Sequence and Tracheal Intubation	High			X	X	X	(X) [§]	X	X	X
Cricothyrotomy	High				X	X		X	X	X
Tube Thoracostomy	High	X	X		X	X	(X)	X	X	X
Thoracocentesis	High	X	X		X		(X)	X	X	X
Thoracotomy	High				X	X		X	X	X
Cardioversion	High			X	X	X	(X)	X	X	X
Pericardiocentesis	High				X	X		X	X	X

Pines JM, Kelly JJ, Meisl H, Augustine JJ, Broida RI, Clarke JR, et al. Procedural safety in emergency care: a conceptual model and recommendations. Jt Comm J Qual Patient Saf 2012; 38:516-526.

Pediatric Emergency Department pre-sedation checklist

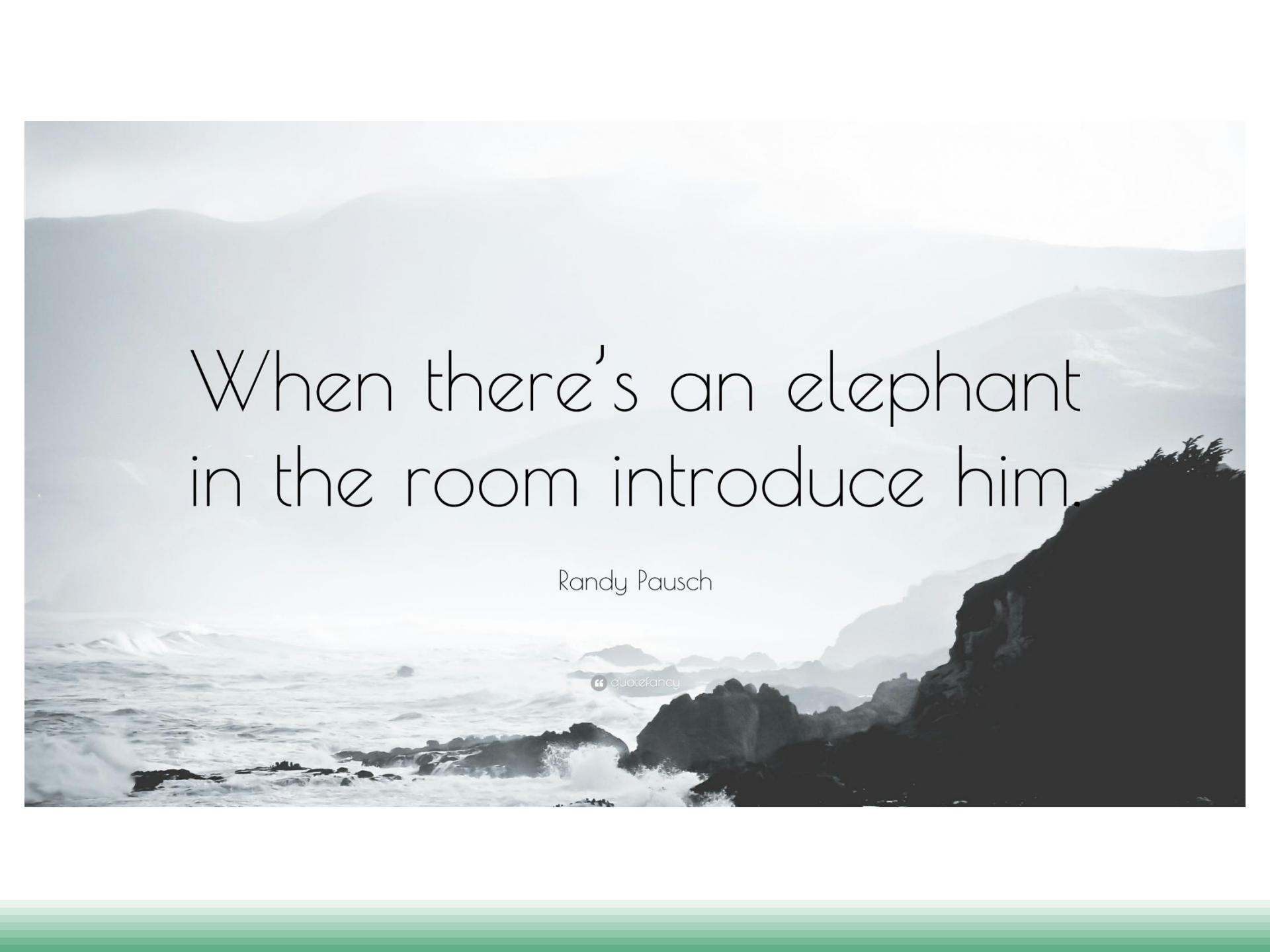


Possible complications

- Airway obstruction (kissing tonsils?)
- Hypoxia and hypoventilation
- Apnea
- Cardiovascular Depression (extremely rare)
- Aspiration (Stomach content, blood)
- Head injury due to a fall

**“I suppose I’ll be the one to mention that there is
an elephant in the emergency room”**





When there's an elephant
in the room introduce him.

Randy Pausch

“ quote fancy

PEDIATRICS Vol. 105 No. 4 April 2000, pp. 805-814
**Adverse Sedation Events in Pediatrics: A Critical
Incident Analysis of Contributing Factors**

DATABASE

FDA adverse drug event reporting system, US Pharmacopeia, a survey of pediatric specialists.

- ▶ 95 Cases of complications
- ▶ 51 death cases, 9 cases of brain damage, 21 cases of prolonged hospitalizations
- ▶ 5 years old children

TABLE 2. Specialty Performing Sedation and Outcome

Specialty	n	%	Outcome	
			Death or Permanent Neurologic Injury n (%)	Prolonged Hospitalization Without Injury or No Harm n (%)
Total dental	32	33.7	29 (91)	3 (9)
Unknown dental specialty	16	16.8	14 (88)	2 (12)
Oral surgery	11	11.6	10 (91)	1 (9)
Pedodontist	3	3.2	3 (100)	0 (0)
General dentist	1	1.0	1 (100)	0 (0)
Dental nurse anesthetist	1	1.0	1 (100)	0 (0)
Unknown medical specialty	19	20.0	8 (42)	11 (58)
Radiology	15	15.8	11 (73)	4 (27)
Cardiology	5	5.3	3 (60)	2 (40)
Oncology	5	5.3	0 (0)	5 (100)
Emergency medicine	4	4.2	0 (0)	4 (100)
Gastroenterology	4	4.2	1 (25)	3 (75)
Unknown pediatric medical	4	4.2	2 (50)	2 (50)
Audiology	2	2.1	1 (50)	1 (50)
Gynecology	2	2.1	2 (100)	0 (0)
General pediatrician	2	2.1	2 (100)	0 (0)
Surgeon	1	1.0	1 (100)	0 (0)
Total	95		60	35

The sedation provider was the “one to blame” (competent provider = safety)

- **Systematic approach from the beginning of sedation until recovery did not exist**
- **Incorrect monitoring of the patient**
- **Lack of resuscitation skills when needed most**

- **Sentinel adverse events** (permanent neurologic injury or death) occurred more frequently in a nonhospital-based facility
- **Successful outcomes** (prolonged hospitalization or no harm) occurred more frequently in a hospital-based setting.
- **Inadequate resuscitation** was more often associated with a nonhospital-based setting.
- Inadequate and inconsistent physiologic monitoring (particularly failure to use or respond appropriately to pulse oximetry) was a major factor contributing to poor outcomes in all venues.
- **Inadequate pre-sedation medical evaluation**
 - Lack of an independent observer
 - Medication errors
 - inadequate recovery procedures

Important points that are essential to safe sedation

- ▶ Pre-sedation evaluation
("Can I sedate this patient?")
- ▶ Preplanning of sedation depth and Length
("is deep sedation or dissociative sedation really needed?")
- ▶ Midazolam sedation = Observing and assessing the patient's vital signs and symptoms
Dissociate/deep sedation = adding a Sat/ECG monitoring or Sat/ETco₂ Monitoring
- ▶ Rescue intervention is always ready
Bag and mask (proper sizes for patient's age) connected to an oxygen supply valve, feeding tube connected to a vacuum pump
- ▶ Use medications that you are familiar with
- ▶ Recording
- ▶ Highly skilled team member available for a hypothetical worse-case-scenario

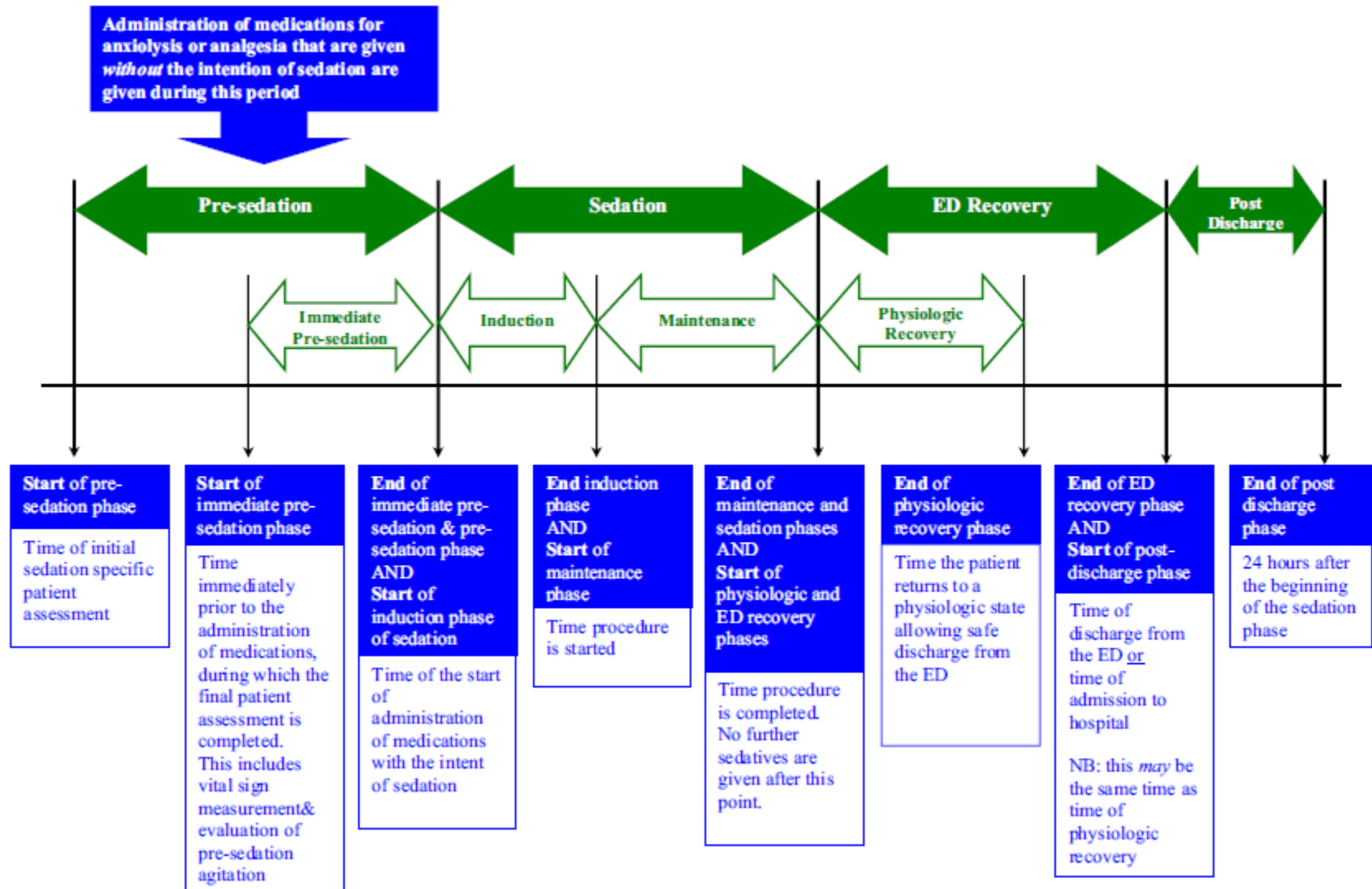
Depth of sedation

The patient's sedation level can be easily changed from minimal sedation to moderate and deep sedation. The patient must be continuously monitored and the sedation provider must be prepared to deal with any possible adverse reaction

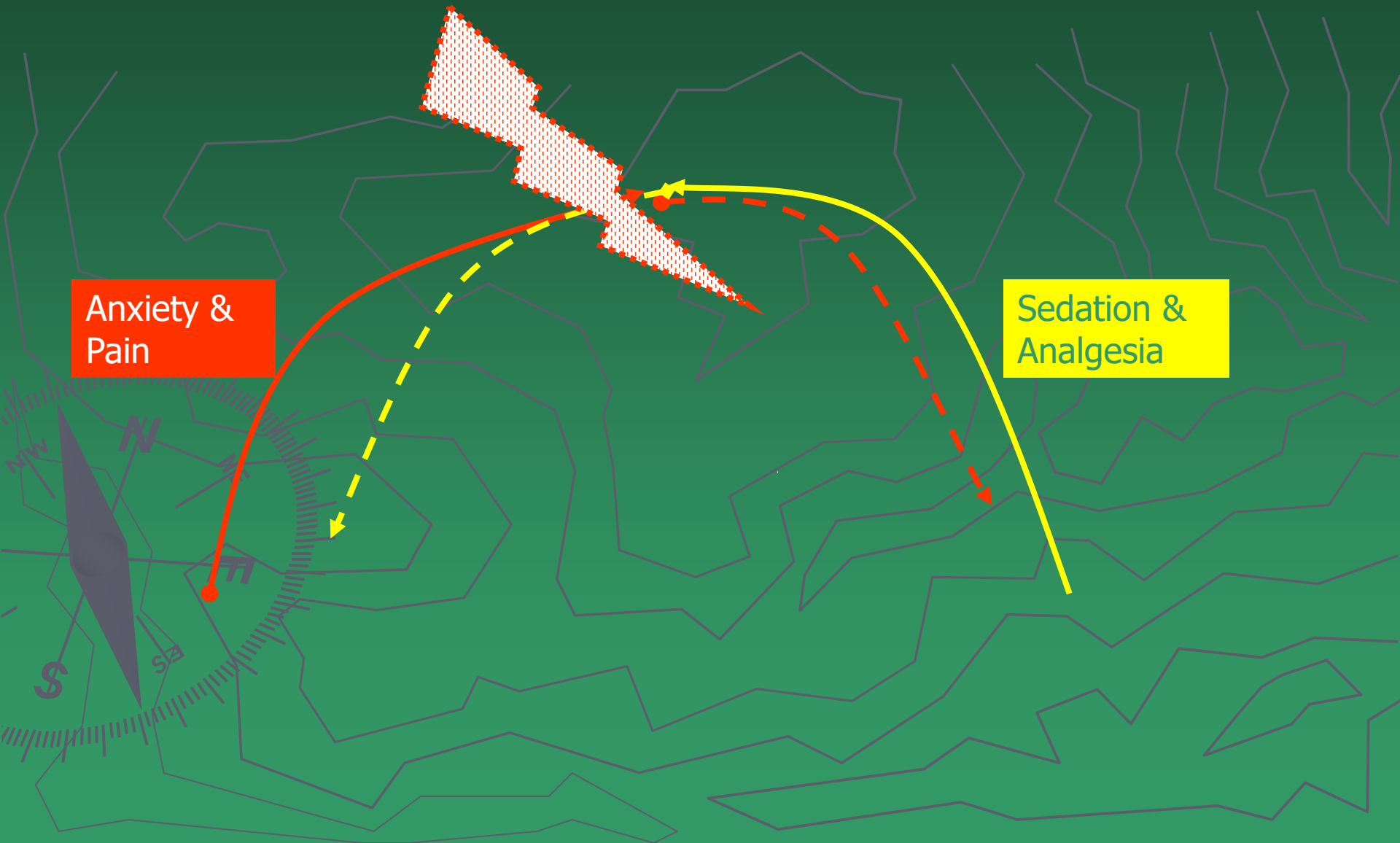
Proper administration of sedation medications

- Use two drugs if one drug doesn't seem to provide satisfactory effect
- Slow titration
- Be familiar with the drugs you use (Propofol is given in a bolus dose of $\sim 1-2$ mg/kg, induction of anesthesia takes ~ 20 s, anesthesia lasts $\sim 5-10$ minutes)
- Do not go over the pharmacologic dose
- Expect adverse events and deal with them

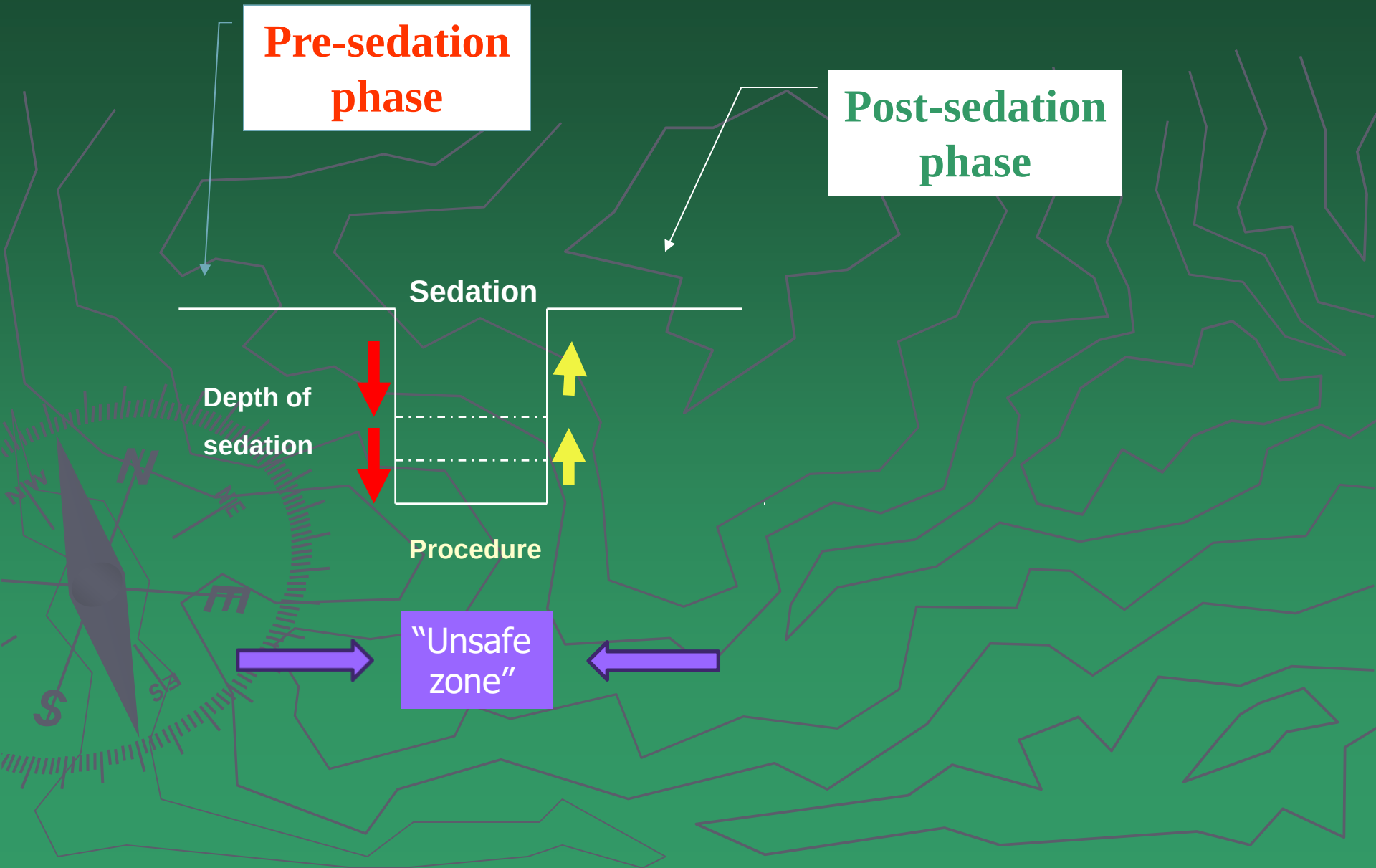
Sedation time intervals



The basic principle



The Ideal scenario



The real-life scenario

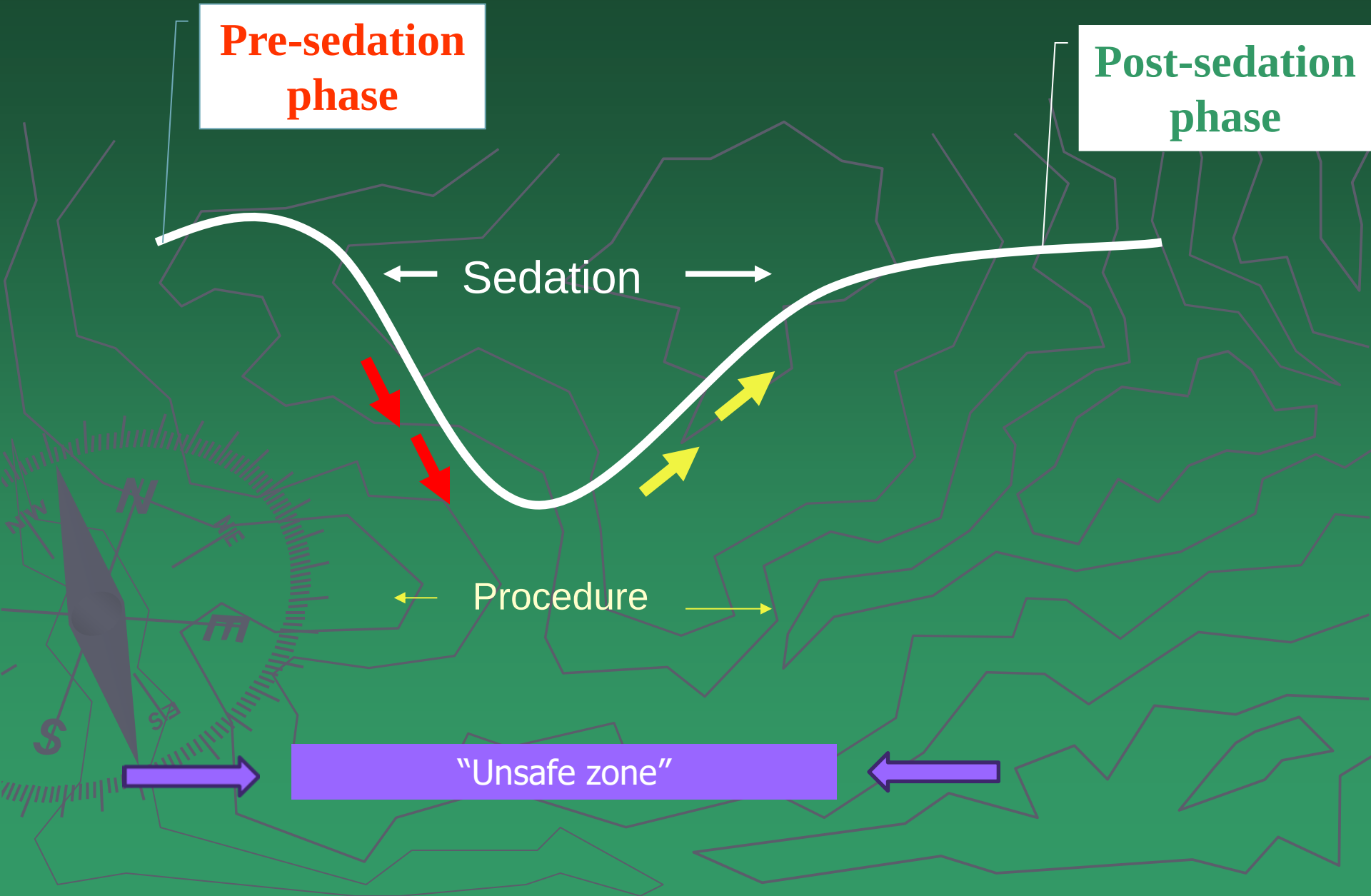
**Pre-sedation
phase**

**Post-sedation
phase**

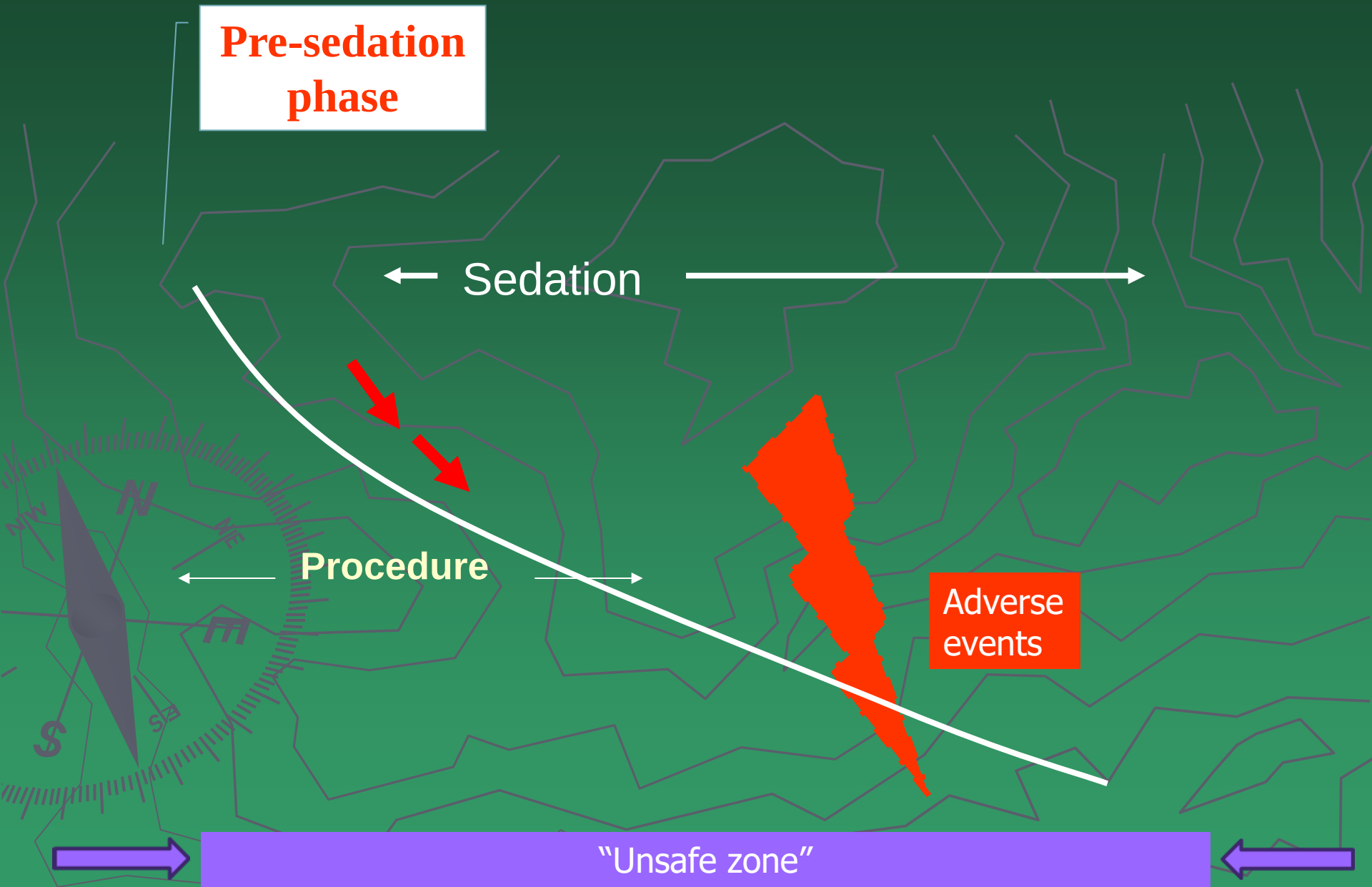
Sedation

Procedure

"Unsafe zone"



The wrong scenario



Discharge criteria

- ▶ Alert and fully conscious
- ▶ Normal heart rate, blood pressure and oxygen saturation
- ▶ Not complaining of pain
- ▶ No vomiting or nausea
- ▶ Can walk steadily
- ▶ Time of the day? Competent caregivers?

The highest-risk stage of the sedation process is...?

After it ends (in recovery phase)